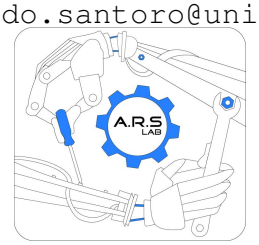
Elements of Model Predictive Control

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Advanced Robotics and Autonomous Systems

Notes

Overview of Control Strategies

Control Strategies

- **PID** simple, step response, single state variabile, linear systems
- **Fuzzy** multi variable, linear and non linear systems
- **LQR** multi variabile, cost function, linear systems

Notes

Model Predictive Control

Given a discrete dynamic system:

$$\begin{aligned} \mathbf{x}(k+1) &= f(\mathbf{x}(k), \mathbf{u}(k)) \\ \mathbf{y}(k) &= g(\mathbf{x}(k), \mathbf{u}(k)) \end{aligned}$$

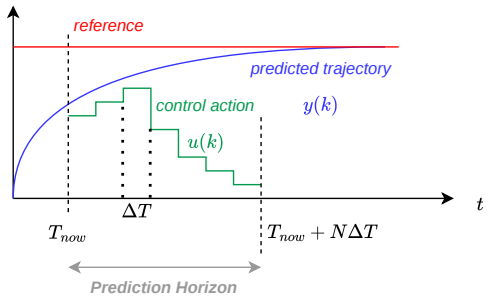
- It is based on computing the **optimal control action** $\mathbf{u}(k+1), \mathbf{u}(k+2), \ldots, \mathbf{u}(k+n)$ to be applied in the near future
- **optimality** is based on a certain **cost function** $J(\mathbf{x}, \mathbf{u})$ to be minimized

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Basics of Model Predictive Control

Receding Horizon

MPC is based on defining a **Prediction Horizon**, i.e. what it will happen from now to the next N steps



MPC — Receding Horizon

For each control step k:

1. Compute the input sequence $\mathbf{u}(k+1), \mathbf{u}(k+2), \ldots, \mathbf{u}(k+N)$ that **minimizes** a cost function:
$$\sum_{i=1}^{N} J(\mathbf{u}(k+i), \mathbf{x}(k+i))$$
2. Apply only the **first result** $\mathbf{u}(k+1)$
3. Let the system evolve (sample time) and repeat the procedure

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MPC – Optimization Problem

MPC requires, at each step, to solve a **constrained optimization problem**:

Find $\mathbf{u}(k+1), \mathbf{u}(k+2), \ldots, \mathbf{u}(k+N)$ such that:

$$\min_{\mathbf{x}(k+i),\mathbf{u}(k+i)} \quad \sum_{i=1}^{N} J(\mathbf{u}(k+i), \mathbf{x}(k+i))$$

$$\textit{subject to} \quad \mathbf{x}(k+1) = f(\mathbf{x}(k), \mathbf{u}(k))$$
$$\mathbf{y}(k) = g(\mathbf{x}(k), \mathbf{u}(k))$$

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Constrained Optimization Problems

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Constrained Optimization Problems

Simple Minimization

Find x such that:

$$\min_{\mathbf{x}} \quad x^2 - 3x - 2$$

`examples/constrainted_optimization/cop_example_1.ipynb`

Notes

Constrained Optimization Problems

Constrained Maximization

Find h and w such that:

$$\begin{aligned} \max_{\mathbf{h},\mathbf{w}} \quad & h \cdot w \\ s.t. \quad & 2(h+w) = 8 \end{aligned}$$

Find h and w of maximum area a rectangle that has the perimeter of 8 meters

`examples/constrainted_optimization/cop_example_2.ipynb`

Notes

Constrained Optimization Problems

Quadratic Constrained Minimization

Find x_0 and x_1 such that:

$$\begin{aligned} \min_{\mathbf{x_0},\mathbf{x_1}} \quad & x_0^2 + x_1^2 - 3x_0x_1 - x_0 - 2x_1 \\ s.t. \quad & x_0 + x_1 \leq 8 \\ & x_0 \geq 0 \\ & x_1 \geq 0 \end{aligned}$$

`examples/constrainted_optimization/cop_example_3.ipynb`

Notes

Standard Quadratic Constrained Minimization

The Quadratic Problem is often defined using matrices:

Find **x** (column vector) such that:

$$\min_{\mathbf{x}} \quad \frac{1}{2}\mathbf{x}^T Q\,\mathbf{x} + c^T\,\mathbf{x}$$

where Q is the symmetric matrix of quadratic coefficients and c is the (column) vector of the linear coefficients

`examples/constrainted_optimization/cop_example_4.ipynb`

Notes

MPC with Linear Quadratic Optimization

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MPC with Linear Quadratic Optimization

Given a discrete dynamic system:

$$\begin{aligned} \mathbf{x}(k+1) &= f(\mathbf{x}(k), \mathbf{u}(k)) \\ \mathbf{y}(k) &= g(\mathbf{x}(k), \mathbf{u}(k)) \end{aligned}$$

Let's define:

- the **horizon** N
- a target state $\overline{\mathbf{x}}$
- an **error cost matrix** Q
- an **input cost matrix** R
- the constraints of the **input:** $\mathbf{u}_{min} \leq \mathbf{u} \leq \mathbf{u}_{max}$

The objective function is:

$$\min_{\mathbf{x},\mathbf{u}} \quad \sum_{i=1}^{N} (\overline{\mathbf{x}} - \mathbf{x}(k+i))^T \, Q \, (\overline{\mathbf{x}} - \mathbf{x}(k+i)) + \mathbf{u}(k+i)^T \, R \, \mathbf{u}(k+i)$$

Notes

Model Predictive Control

MPC Pros

- For each step a new optimal solution is computed
- It can be applied to linear and non-linear systems
- Constrains can be specified

MPC Cons

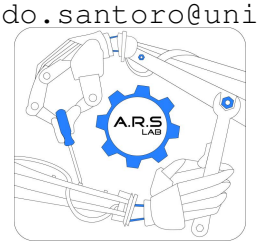
- An optimization problem must be solved for each step
- High computational cost

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